Yonghui Liu

Machine Learning Research & Software Engineering

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TECHNICAL SKILLS

Tools: Python, PyTorch, TypeScript, React, FastAPI, SQLite.

Methods: Sequence modeling, graph representation learning, language-model supervision, conditional generation.

EXPERIENCE

Doctoral Research | KAIST Aug 2023 - Present

- Developed Transformer-based trajectory reconstruction and evaluated the contributions of different inputs to motion prediction.
- Developed road-network representations and label-alignment methods using language models and graph learning, supporting cross-city transfer and learning with limited labels.

Software Engineer | Meituan May 2021 - Aug 2023

AI Research Intern Apr 2020 - Oct 2020

Advanced Institute of Information Technology, Peking University

PROJECTS

Generative Traffic Simulation Ongoing research

- Developing route-conditioned traffic generation with joint model fine-tuning for autonomous-driving simulation.

Adaptive Sensing Web (ATSW) Research prototype

- Built a Python/FastAPI and React/TypeScript system for GPS replay, adaptive reporting, and receiver-state estimation.

EDUCATION

KAIST | Ph.D. candidate Aug 2023 - Present

Cho Chun Shik Graduate School of Mobility

Zhejiang University | M.S. in Ocean Engineering Sep 2018 - Mar 2021

Nanchang University | B.S. in Physics Sep 2014 - May 2018

PUBLICATIONS

Four first-author papers in Knowledge-Based Systems, Communications in Transportation Research, and IEEE ITSC (2025-2026).